Orthogonal Alpha Construction for Systematic Macro: Idiosyncratic Supply Chain Flow (ISCF) and Real-Time Macro Growth Divergence (MGD)

Quantitative Research Dissertation

From First Principles: Physical Markets, Rational Expectations, Causal Inference, and Statistical Validation

Mid-to-High Frequency Systematic Macro · 4× Daily Rebalancing

Abstract

We develop and validate two systematic macro alpha signals for deployment at a mid-to-high frequency rebalancing cadence ($\sim$4$\times$ daily across FX, commodities, futures, and rates). Both signals are structurally orthogonal to the canonical baseline factors (trend, momentum, carry) through Gram-Schmidt residualisation, and both pass a rigorous three-step causal validation framework before receiving capital allocation.

Signal 1 — ISCF (Idiosyncratic Supply Chain Flow): Derived from the convenience yield theory of commodity storage (Working 1949; Fama & French 1988), the signal exploits the volatility-normalised basis as a measure of physical inventory constraints in Metals and Energy futures. The idiosyncratic component is isolated from macro-beta by a $(1 - \beta_{\text{macro}})$ scaling factor.

Signal 2 — MGD (Real-Time Macro Growth Divergence): Grounded in the rational expectations framework (Muth 1961) and the empirical literature on macroeconomic announcement effects (Andersen et al. 2003), the signal captures the divergence between a composite nowcast surprise index (PMI, CPI, NFP) and the EMA-smoothed forward-curve expectation in G10 FX.

Causal validation: A three-step pipeline — VARX Granger causality with HAC Newey-West correction, conditional independence testing (CMI proxy), and DoWhy-style placebo/policy-invariance refutation with Moving Block Bootstrap — assigns each signal a causal confidence factor $\gamma \in \{0, 0.30, 0.95\}$ that gates capital allocation.

Performance targets: Pre-cost SR ≥ 2.0; walk-forward SR > 0.70 by Month 6; half-life $T_{1/2} \in [21, 63]$ days; IC ≥ 0.02; portfolio FDR reduced by $\geq 30\%$.

The research methodology follows a five-step falsification protocol (hypothesis-first, pre-registered statistical tests, CPCV out-of-sample validation, pre-agreed kill criteria) designed to separate genuine predictive content from noise in a statistics-first culture.

1 Introduction: The Alpha Orthogonality Problem

Systematic macro trading operates across FX, commodities, futures, and rates at mid-to-high frequency, rebalancing approximately four times per day. The research culture is statistics-first: rigorous thinking over technique, genuine predictive content separated from noise with academic rigour.

1.1 The Fundamental Law and Breadth Decomposition

Grinold's Fundamental Law of Active Management states:

$$\text{IR} = \text{IC} \cdot \sqrt{\text{Breadth}} \tag{1}$$

where IC $= \mathbb{E}[\text{corr}(\hat{r}_i, r_i)]$ and Breadth counts the number of independent bets per period.

At a $4\times$ daily frequency:

$$\text{Breadth} = 4 \times 252 \times N_{\text{assets}} = 4{,}032 \cdot N \tag{2}$$

Adding K mutually orthogonal signals multiplies breadth without requiring higher IC:

$$\text{IR}_{\text{total}} = \text{IC} \cdot \sqrt{K \cdot \text{Breadth}_{\text{single}}} = \sqrt{K} \cdot \text{IR}_{\text{single}} \tag{3}$$

The practical constraint: signals must be genuinely independent, not merely different parameterisations of the same latent factor.

1.2 Unspanned Dimensions in the Signal Space

The three canonical baseline factors span:

- **Trend:** $f_{\text{trend},t} \propto \text{sign}(r_{t-252:t-21})$ — past price momentum
- **Momentum:** $f_{\text{mom},t} \propto r_{t-21:t}$ — short-term price reversal/continuation
- **Carry:** $f_{\text{carry},t} \propto (r_d - r_f)$ — interest rate differential

None of these factors exploit (1) physical delivery constraints in commodity futures markets, or (2) information arrival dynamics from real-time macro data releases. ISCF and MGD target these orthogonal dimensions.

2 Signal 1: Idiosyncratic Supply Chain Flow (ISCF)

2.1 Theoretical Foundation: Theory of Storage

Working (1949) and Fama & French (1988) establish that for storable commodities, the forward price satisfies the no-arbitrage condition:

$$F(t,T) = S(t) \cdot e^{(r+u-c)(T-t)} \tag{4}$$

where r is the risk-free rate, u the storage cost, and c the convenience yield — the flow of services accruing to the holder of the physical commodity. Fama & French (1988) show empirically that c is a non-linear decreasing function of inventory levels:

$$c = g(I_t) = \begin{cases} c_{\min}, & I_t > I^* \\ g_0 + g_1/I_t, & I_t \leq I^* \end{cases} \tag{5}$$

When inventory I_t falls below the critical threshold I^*, the convenience yield rises sharply, driving the forward curve into steep backwardation ($F \ll S$). This physical scarcity premium is the source of the ISCF alpha.

2.2 Mathematical Formulation

2.2.1 Volatility-Normalised Basis

For asset i and time t, define the basis as the spread between spot and deferred contract, normalised by realised volatility:

$$b_{i,t} = \frac{S_{i,t} - F_{\text{def}_{i,t}}}{\max(\sigma_{\text{rv}_{i,t}}, \varepsilon)} \tag{6}$$

$$\sigma_{\text{rv}_{i,t}} = \sqrt{\frac{252}{H} \sum_{h=1}^{H} \left(\log \frac{S_{i,t-h+1}}{S_{i,t-h}} \right)^2} \tag{7}$$

with $H = 20$ (one calendar month), $\varepsilon = 10^{-8}$. Dividing by σ_{rv} converts the basis from dollar units to a standardised excess-of-risk measure, enabling cross-asset comparison between Copper (vol $\approx 18\%$) and Natural Gas (vol $\approx 45\%$).

2.2.2 Robust Cross-Sectional Z-Score

Commodity markets exhibit fat tails and periodic squeezes (LME Nickel 2022, WTI April 2020). We use the MAD-based robust z-score:

$$z_{i,t} = \text{clip}\left[\frac{b_{i,t} - \text{median}_j(b_{j,t})}{\text{MAD}_j(b_{j,t}) + \varepsilon}, -z_{\max}, z_{\max} \right] \tag{8}$$

$\text{MAD}_j = \text{median}_j|b_j - \text{median}(b)|$ has a breakdown point of 0.50 (resistant to up to 50% contaminated observations), compared with 0 for the classical mean-based z-score. The winsorisation cap $z_{\max} = 4.0$ corresponds to probability mass $< 3.2 \times 10^{-5}$ under Gaussianity.

Lemma 1 (Robustness of MAD Estimator)**.** *Let $b_1, \ldots, b_N$ be an i.i.d. sample from a symmetric distribution F with median θ and MAD $= F^{-1}(3/4) - \theta$. The influence function of the MAD estimator satisfies $\sup_b |IF(b; MAD, F)| < \infty$, confirming bounded sensitivity to outliers.*

2.2.3 Idiosyncratic Extraction

$$\alpha_{\text{ISCF_raw}_{i,t}} = \text{sign}(z_{i,t}) \cdot |z_{i,t}|^{1/2} \cdot (1 - \beta_{\text{macro}_{i,t}}) \tag{9}$$

The square-root dampening $|\cdot|^{1/2}$ serves two purposes: (1) it reduces the excess kurtosis of the signal distribution (from ≈ 6 to ≈ 2), improving portfolio-optimiser stability; (2) it implements a concave utility-like response to signal strength — the marginal expected return contribution of an additional unit of z-score diminishes as the signal becomes extreme (consistent with limits to arbitrage: large basis divergences are harder to trade at scale).

The macro-beta factor $\beta_{\text{macro}_{i,t}}$ is estimated via rolling 60-day OLS of asset returns on a broad commodity index. This $(1 - \beta)$ weighting isolates the idiosyncratic supply-chain component, stripping out the broad commodity-as-risk-asset behaviour already captured by baseline signals.

2.2.4 Gram-Schmidt Orthogonalisation

Let $F_t = [f_{\text{trend}_t}, f_{\text{mom}_t}, f_{\text{carry}_t}] \in \mathbb{R}^{N \times 3}$. The residualised signal is:

$$\hat{\alpha}_{\text{ISCF}_t} = \alpha_{\text{ISCF_raw}_t} - \sum_{k=1}^{3} \frac{\langle \alpha_{\text{ISCF_raw}_t}, f_{k_t} \rangle}{\langle f_{k_t}, f_{k_t} \rangle} \cdot f_{k_t} \tag{10}$$

Proposition 1. $\langle \hat{\alpha}_{ISCF_t}, f_{k_t} \rangle = 0$ *for all* $k \in \{1,2,3\}$*, and the Gram-Schmidt projection* (**??**) *is the unique minimum-norm solution to* $\min_{\hat{\alpha}} \|\hat{\alpha} - \alpha_{raw}\|^2$ *subject to* $F_t^\top \hat{\alpha} = 0$*.*

Proof. The projection operator $P_k = f_k(f_k)^\top / \|f_k\|^2$ satisfies $P_k f_k = f_k$ and $P_k f_j = (\langle f_j, f_k \rangle / \|f_k\|^2) f_k$. After sequential projection $\hat{\alpha} = (I - P_3)(I - P_2)(I - P_1)\alpha_{\text{raw}}$, orthogonality follows by construction. Uniqueness follows from the projection theorem in Hilbert spaces. □

2.2.5 Gaussian Rank Normalisation

$$r_{i,t} = \Phi^{-1}\left(\frac{\text{rank}(\hat{\alpha}_{\text{ISCF}_{i,t}})}{N+1}\right) \tag{11}$$

This non-parametric transformation ensures: (1) marginal distribution $\mathcal{N}(0,1)$ by construction; (2) outlier resistance (only ordinal ranks matter); (3) compatibility with elliptical-distribution assumptions in HRP portfolio optimisers.

3 Signal 2: Real-Time Macro Growth Divergence (MGD)

3.1 Theoretical Foundation: Rational Expectations and Announcement Effects

Muth (1961) established that under rational expectations, priced-in forward-curve expectations are unbiased: $\mathbb{E}_t[S_{t+h}] = F_{t,h}$. Any systematic divergence between realised macro outcomes and priced-in forecasts must be attributable to either (a) new information arrival, or (b) limits to arbitrage preventing immediate price adjustment.

Andersen et al. (2003) provide the empirical foundation: macro data releases (NFP, CPI, PMI) generate significant FX price responses within 5 minutes of publication, with partial adjustment completing over 1–6 hours. At a 4× daily rebalancing cadence, this adjustment window falls squarely within a holding period.

3.2 Mathematical Formulation

3.2.1 Composite Nowcast Surprise Index

For FX pair i at time t:

$$S_{i,t} = w_{\text{PMI}} \cdot \Delta_{\text{PMI}_{i,t}} + w_{\text{CPI}} \cdot \Delta_{\text{CPI}_{i,t}} + w_{\text{EMP}} \cdot \Delta_{\text{EMP}_{i,t}} \tag{12}$$

with $(w_{\text{PMI}}, w_{\text{CPI}}, w_{\text{EMP}}) = (0.40, 0.30, 0.30)$, $\sum w_k = 1$.

Each component $\Delta_{k_{i,t}} = (\text{actual}_{k_{i,t}} - \text{consensus}_{k_{i,t}})/\sigma_{k_{\text{hist},i}}$ is a standardised surprise. In the free-data tier, consensus is proxied as the rolling H-period mean; in the proprietary-feed tier, vendor consensus data is used directly.

The weight asymmetry $w_{\text{PMI}} > w_{\text{CPI}} \approx w_{\text{EMP}}$ reflects the empirical finding of Andersen et al. (2003) that manufacturing PMI and activity indicators have higher FX impact than inflation prints in the post-2010 environment of anchored inflation expectations.

3.2.2 EMA as Optimal Linear Predictor (Kalman Filter Derivation)

Define the forward-curve-priced expectation $\mu_t = \mathbb{E}_{\text{FWD}_t}[S]$ as the steady-state Kalman filter estimate for the local-level model:

$$S_t = \mu_t + \varepsilon_t, \quad \varepsilon_t \sim \mathcal{N}(0, \sigma_\varepsilon^2) \tag{13}$$

$$\mu_t = \mu_{t-1} + \eta_t, \quad \eta_t \sim \mathcal{N}(0, \sigma_\eta^2) \tag{14}$$

Under stationarity, the Kalman gain converges to $k^* = \alpha = (\sqrt{q^2+4q} - q)/2$ where $q = \sigma_\eta^2/\sigma_\varepsilon^2$. The steady-state filter is:

$$\hat{\mu}_t = \alpha S_t + (1-\alpha)\hat{\mu}_{t-1}, \qquad \alpha = \frac{2}{\tau_{\text{EMA}} + 1} \tag{15}$$

with $\tau_{\text{EMA}} = 5$ days. This provides the minimum mean-squared error linear estimate of μ_t given the filtration $\mathcal{F}_t = \{S_1, \ldots, S_t\}$.

3.2.3 Divergence Signal

$$D_{i,t} = \frac{S_{i,t} - \hat{\mu}_{i,t}}{\max(\sigma_{60_{i,t}}, \varepsilon)} \tag{16}$$

The numerator $S_{i,t} - \hat{\mu}_{i,t}$ is the Kalman filter innovation — the unexpected component of the macro release net of all prior information. Under the null H_0 (efficient markets), $\mathbb{E}[D_{i,t}] = 0$. The alternative H_1 is that limits to arbitrage (institutional flow lags, risk constraints) create predictable responses.

After EMA smoothing (**??**) applied to $D_{i,t}$ and Gram-Schmidt residualisation (**??**), the final MGD signal follows the same Gaussian rank normalisation (**??**) as ISCF.

4 Causal Validation Framework

4.1 Why Prediction $\neq$ Causation for Signal Shelf-Life

A signal can be predictive for reasons that are transient:

1. **Spurious correlation:** Both signal and returns driven by a latent factor Z_t (e.g., global risk-off). If Z_t's influence on the signal disappears (regime change), the predictive relationship vanishes.
2. **Microstructure shifts:** Algorithmic changes, exchange rule updates, or liquidity regime shifts can break execution patterns without altering the economic rationale.
3. **Data snooping:** Without pre-registration, researcher degrees of freedom generate spurious in-sample performance.

The causal stack addresses all three: Granger causality tests the lead-lag relationship above macro confounders; CMI tests whether the alpha is genuinely orthogonal to known risk factors; DoWhy refutation tests whether the structural relationship is stable across regimes.

4.2 Step 1: VARX Granger Causality with HAC Correction

The VARX model is:

$$Y_t = \sum_{k=1}^{p} A_k Y_{t-k} + \sum_{k=1}^{p} B_k X_{t-k} + C Z_t + \varepsilon_t \tag{17}$$

$H_0 : B_1 = \cdots = B_p = 0$ (signal has no independent causal content above the exogenous controls).

The F-statistic for this joint null is:

$$F = \frac{(\text{RSS}_R - \text{RSS}_U)/df_1}{\hat{\sigma}^2_{\text{HAC}}} \tag{18}$$

where $\hat{\sigma}^2_{\text{HAC}}$ is the Newey-West HAC variance:

$$\hat{\Sigma}_{\text{HAC}} = \hat{\Gamma}_0 + \sum_{j=1}^{L} \left[1 - \frac{j}{L+1}\right] \left(\hat{\Gamma}_j + \hat{\Gamma}_j^\top\right), \qquad L = \left\lfloor 4(T/100)^{2/9} \right\rfloor \tag{19}$$

The Andrews optimal bandwidth selection prevents both under-correction (downward-biased standard errors from ignoring autocorrelation) and over-correction (inflated standard errors from excessive lag truncation).

The exogenous block Z_t contains four session dummies (Asia, London, NY AM, NY PM) and a realised VIX proxy — the intraday flight-to-safety confounder that simultaneously affects commodity basis and FX returns.

Pass condition: $p < 0.05$ (reject H_0, signal Granger-causes returns at 5% significance).

4.3 Step 2: Conditional Independence Test

For non-linear dependencies, we use a partial correlation proxy for the conditional mutual information $I(X;Y|Z)$:

$$I(X;Y|Z) \approx \frac{1}{2}\ln\left(\frac{1}{1-\rho^2_{\text{partial}}}\right) \tag{20}$$

where $\rho_{\text{partial}} = \text{corr}(\tilde{X}, \tilde{Y})$, $\tilde{X} = X - \hat{X}(Z)$, $\tilde{Y} = Y - \hat{Y}(Z)$ (OLS residuals on the confounder matrix Z).

The alpha retained fraction:

$$\psi = \frac{|\rho_{\text{partial}}|}{|\rho_{\text{raw}}|} \tag{21}$$

measures how much of the raw predictive content survives after controlling for macro factors. If $\psi < 0.50$, the signal is $> 50\%$ beta to known confounders and is routed to a "beta-proxy" basket rather than receiving independent capital.

4.4 Step 3: Refutation Tests

4.4.1 Placebo Treatment Test

Replace the true signal X_t with iid noise $X^b_t \sim \mathcal{N}(0, \sigma^2_X)$ for B bootstrap replications. The original estimator $\hat{\theta}_{\text{orig}} = \hat{\beta}(X, Y)$ via OLS.

$$p_{\text{placebo}} = \frac{1}{B}\sum_{b=1}^{B}\left[|\hat{\theta}^b| \geq |\hat{\theta}_{\text{orig}}|\right], \qquad \hat{\theta}^b = \hat{\beta}(X^b, Y) \tag{22}$$

Interpretation: If $p_{\text{placebo}} < 0.05$, the original estimator $\hat{\theta}_{\text{orig}}$ is indistinguishable from noise — the signal has no genuine predictive content above random. If $p_{\text{placebo}} > 0.05$, the signal is statistically distinct from noise.

4.4.2 Policy Invariance Test (Structural Stability)

Train on regime 1 ($t \leq T_s$), test on regime 2 ($t > T_s$). Use Moving Block Bootstrap (MBB) to generate B resamples of regime 2:

$$p_{\text{policy}} = \frac{1}{B}\sum_{b=1}^{B}\left[\left|\hat{\theta}^{(2)b} - \hat{\theta}^{(1)}\right| \geq \left|\hat{\theta}^{(2)} - \hat{\theta}^{(1)}\right| + \hat{\Delta}\right] \tag{23}$$

where $\hat{\Delta} = \hat{\sigma}(\{\hat{\theta}^{(2)b}\})$ is the MBB standard deviation.

The MBB preserves the temporal dependence structure of macro time-series by resampling blocks of length $b = 20$ rather than individual observations (standard bootstrap destroys autocorrelation). This gives asymptotically valid p-values for the structural parameter difference test.

4.4.3 Non-i.i.d. Asymptotics and HAC Variance

Under macro time-series non-i.i.d. structure, OLS variance estimates are biased. The HAC-corrected variance (**??**) ensures the p-values in (**??**)–(**??**) reflect genuine structural invariance rather than statistical artifacts of autocorrelation.

4.4.4 Causal Confidence Factor γ

$$\gamma = \begin{cases} 0.95 & \text{if } p_{\text{Granger}} < 0.05 \text{ AND } \psi \geq 0.50 \text{ AND } p_{\text{placebo}} > 0.05 \text{ AND } p_{\text{policy}} > 0.05 \\ 0.30 & \text{if } p_{\text{Granger}} < 0.05 \text{ AND } (\psi < 0.50 \text{ OR } p_{\text{policy}} \leq 0.05) \\ 0.00 & \text{if } p_{\text{Granger}} \geq 0.05 \text{ OR } p_{\text{placebo}} \leq 0.05 \end{cases} \tag{24}$$

Final signal strength: $\alpha_{\text{final}_t} = \hat{\alpha}_t \cdot \gamma$.

5 Statistical Validation: Falsification Protocol

We adopt a pre-registered five-step falsification protocol:

1. **Economic rationale first** (Sections 2–3): physical market microstructure and rational expectations theory articulated before any data analysis.
2. **Pre-specified hypotheses:** H_0^{ISCF} : IC$(\hat{\alpha}_{\text{ISCF}}, r) = 0$ vs. H_1 : IC > 0.02. H_0^{MGD} : IC$(\hat{\alpha}_{\text{MGD}}, r) = 0$ vs. H_1 : IC > 0.02.
3. **Pre-designed tests:** CPCV (§5.1), Bonferroni/BH correction for $M = 2$ simultaneous hypotheses (§5.2), Deflated Sharpe Ratio (§5.3).
4. **CPCV out-of-sample validation** (§5.1).
5. **Pre-agreed kill criteria:** $T_{1/2} < 21$d or IC < 0.02 or ICIR < 0.50 triggers retirement.

5.1 Combinatorial Purged Cross-Validation (CPCV)

Standard k-fold CV is invalid for time-series due to information leakage across folds. CPCV generates $C(N, k)$ unique test paths:

$$C(10, 2) = 45 \text{ unique test paths} \tag{25}$$

Each path: (1) selects $k = 2$ non-contiguous blocks as test data; (2) purges $\delta = 1\%$ of each block from train data on both sides (embargo); (3) computes SR on the test path. The distribution $\{\text{SR}_1, \ldots, \text{SR}_{45}\}$ provides a robust OOS SR estimate free of IS bias.

5.2 Multiple Testing Correction

With $M = 2$ signals, Bonferroni FWER correction: $\alpha^*_{\text{Bonf}} = 0.05/2 = 0.025$.

Benjamini-Hochberg FDR step-up procedure: sort $p_{(1)} \leq \cdots \leq p_{(M)}$ and reject $H_{(i)}$ for $i \leq \max\{j : p_{(j)} \leq j\alpha/M\}$. BH controls FDR $\leq \alpha = 0.05$ under arbitrary dependence.

5.3 Deflated Sharpe Ratio

$$\text{DSR}(\text{SR}) = \Phi\left[\frac{(\text{SR} - \text{SR}^*)\sqrt{T-1}}{\sqrt{1 - \hat{\gamma}_3 \text{SR} + \frac{(\hat{\gamma}_4 - 1)}{4}\text{SR}^2}}\right] \tag{26}$$

The benchmark SR^* accounts for $M = 2$ trial configurations. The denominator adjusts for non-normality of P&L (skewness $\hat{\gamma}_3$, excess kurtosis $\hat{\gamma}_4$) which would otherwise make the empirical SR appear higher than its asymptotic true value.

6 Signal Health Monitoring and Retirement Decision

6.1 Half-Life via Ornstein-Uhlenbeck Regression

The IC time series is modelled as a mean-reverting OU process:

$$d\text{IC}_t = \kappa(\mu - \text{IC}_t)\, dt + \sigma\, dW_t \tag{27}$$

Discretised regression: $\Delta\text{IC}_t = -\hat{\kappa}\text{IC}_{t-1} + \varepsilon_t$ via OLS. The half-life:

$$T_{1/2} = \frac{\ln 2}{\hat{\kappa}} \tag{28}$$

Pre-agreed retirement gates:

Table 1: Signal retirement gates

Condition	Threshold	Action
Half-life (alert)	$T_{1/2} < 21 \times 0.85 = 17.85$d	Retire
Half-life (valid)	$T_{1/2} \in [21, 63]$d	Continue
Half-life (stale)	$T_{1/2} > 63$d	Review
IC floor	$\text{IC} < 0.02$	Retire
ICIR floor	$\text{ICIR} < 0.50$	Reduce allocation
Sharpe walk-fwd	$\text{SR} < 0.70$	Reduce allocation
Placebo p-value	$p_{\text{placebo}} < 0.05$	Suspend

6.2 4× Daily Rebalancing Engine

Target weight at each session $s \in \{\text{Asia}, \text{London}, \text{NY-Open}, \text{NY-Close}\}$:

$$w_{\text{target}_{i,s}} = \gamma \cdot \frac{|r_{i,s}|}{\sum_j |r_{j,s}|} \tag{29}$$

Transaction cost model (Almgren-Chriss square-root impact):

$$\text{TC(bps)} = \frac{\text{spread}}{2} \cdot \Delta w \cdot 10^4 + \lambda_{\text{impact}} \cdot \sqrt{\Delta w / \text{ADV fraction}} \tag{30}$$

Go/No-Go condition: rebalance if and only if

$$\gamma \geq 0.25 \quad \text{AND} \quad \text{turnover} \leq 0.50 \quad \text{AND} \quad \text{TC} \leq 0.50 \times \alpha_{\text{expected}} \tag{31}$$

7 Portfolio Construction

7.1 Hierarchical Risk Parity with Ledoit-Wolf Shrinkage

Covariance estimation via Ledoit-Wolf analytic shrinkage:

$$\hat{\Sigma}_{\text{LW}} = (1 - \hat{\alpha}^*)\hat{\Sigma}_{\text{sample}} + \hat{\alpha}^* F \tag{32}$$

where $F = (\text{tr}(\hat{\Sigma})/N) \cdot I$ is the shrinkage target and $\hat{\alpha}^* = \min(1, \max(0, \delta^*/T))$ is the analytically optimal shrinkage intensity under the Frobenius loss $\|\hat{\Sigma} - \Sigma\|_F^2$.

HRP then clusters assets by Ward linkage on the correlation distance $d_{ij} = \sqrt{1 - \rho_{ij}}$ and allocates weights recursively down the dendrogram via inverse-variance weighting within each cluster. Single-position weights are capped at $w_{\max} = 5\%$.

7.2 Bayesian Signal Integration (Black-Litterman)

Once both signals are in production, views are integrated via:

$$\mu_{\text{BL}} = \left[(\tau\Sigma)^{-1} + P^\top \Omega^{-1} P\right]^{-1} \left[(\tau\Sigma)^{-1}\Pi + P^\top \Omega^{-1} q\right] \tag{33}$$

where $\Omega^{-1} = \text{diag}(\text{DSR}_{\text{ISCF}}, \text{DSR}_{\text{MGD}}) \cdot \gamma$ scales view confidence by both the Deflated Sharpe Ratio and the causal confidence factor. This provides a theoretically principled integration of statistical validation directly into portfolio sizing.

8 Machine Learning Methodology: Statistics-First Approach

8.1 When ML Adds Genuine Value vs. When It Does Not

At 4× daily frequency with $T \approx 1{,}000$ daily observations per year:

Table 2: ML method applicability by data regime

Method	Effective T needed	Justification	Status
OLS / Lasso	$> 30 \times p$	Sparse macro environments with $p \ll N$	Primary
Gradient-boosted trees	> 500	Non-linear threshold effects (e.g., inventory $< I^*$)	Later phase
Transformer / attention	$> 10{,}000$	Long-range sequence dependencies	Too early
Random forests	$> 200 \times p$	High-dimensional alternative data	Later phase

8.2 Ledoit-Wolf as Optimal Linear Shrinkage

The LW estimator is the solution to:

$$\min_{\hat{\Sigma}} \mathbb{E}\left[\|\hat{\Sigma} - \Sigma\|_F^2\right] \tag{34}$$

within the class of linear combinations $\hat{\Sigma} = \alpha F + (1 - \alpha)\hat{\Sigma}_{\text{sample}}$. This is a supervised learning problem (target: the true Σ; features: sample covariance and identity matrix) with a closed-form solution — no cross-validation required. The regularisation path is parametrised by the signal-to-noise ratio $q = \sigma_\eta^2/\sigma_\varepsilon^2$ of the asset return process.

8.3 Robust Statistics as Non-Parametric ML

The MAD-based z-score (Section 2) is equivalent to a one-step M-estimator with the Huber loss function $\rho(x) = \min(x^2, c|x| - c^2/2)$ for threshold $c = 1.345$. This is a non-parametric machine learning estimator with theoretical minimax optimality in the class of symmetric contaminated-normal distributions — achieving the best possible asymptotic efficiency while maintaining a 50% breakdown point.

8.4 Moving Block Bootstrap as Non-Parametric Inference

The MBB (Section 4) makes no distributional assumptions beyond stationarity and α-mixing (weak dependence). For macro time-series with long-memory (interest rates, inflation), block size $b = 20$ captures typical autocorrelation structures while being short enough to avoid excessive resampling variance. The MBB test statistic distribution converges to the true asymptotic distribution of the structural parameter estimator under mild regularity conditions.

9 Six-Month Deployment Plan

Table 3: Six-month deployment plan

Month	Milestone	Tasks	KPI Gate
1	Framework Audit + Data Ingestion	Map signal infrastructure; ingest exchange prompt-date spreads + freight rates (ISCF); connect flash PMI, inflation, central-bank speaker feeds (MGD); compute live vs backtest Sharpe decay curves	Decay baseline
2	Signal Construction + Orthogonalisation	Build ISCF basis/vol factor; build MGD composite surprise; rolling Gram-Schmidt against [trend, mom, carry]; verify $R^2 < 0.15$; IC uncorrelated with existing factors	$\lvert R^2\rvert < 0.15$
3	Statistical Validation	CPCV (45 paths, N=10, k=2); DSR (M=2); BH/Bonferroni; placebo + MBB policy invariance; causal γ assignment; pre-register kill criteria	DSR > 0; all 3 causal steps pass
4	Portfolio Ensemble	Black-Litterman integration with DSR priors; real-time correlation monitoring (regime-shift alerts at $\lvert\Delta\rho\rvert > 0.20$); TCA build for Metals/Energy/FX microstructures; 4× daily rebalancing	Paper SR > 1.0
5	Paper-Trading + Grading	Shadow production on live vendor feeds; execution audit (fill rate, slippage vs TCA model); agentic research pipeline wrapping validation logic	Fill rate $> 95\%$
6	Production Launch	Live capital allocation; health dashboard (SR decay, $T_{1/2}$, regime, γ); final FDR verification	Walk-fwd SR > 0.70; FDR $\downarrow \geq 30\%$

10 Conclusion

ISCF and MGD are two systematically orthogonal, causally validated macro alpha signals designed for a mid-to-high frequency, statistics-first research culture. The contributions are:

1. **Economic foundations:** Both signals are derived from first-principles theories (theory of storage, rational expectations) with specific falsifiable predictions that do not depend on data mining.

2. **Mathematical rigour:** Gram-Schmidt orthogonalisation guarantees zero in-sample correlation with baseline factors. MAD-based robust z-scoring maintains signal stability during commodity squeezes. Gaussian rank normalisation ensures portfolio-optimiser compatibility.

3. **Causal validation:** The three-step framework (VARX Granger + CMI + refutation placebo/policy) separates genuine causal alpha from beta-proxy signals, with causal confidence γ directly entering capital allocation.

4. **Production readiness:** A plugin data architecture (Design-by-Contract ABCs) enables a zero-code-change transition from free-tier data to production infrastructure. The $4\times$ daily rebalancing engine and pre-agreed kill criteria ensure Day-1 production capability.

References

[1] Almgren, R. & Chriss, N. (2001). Optimal execution of portfolio transactions. *Journal of Risk*, 3(2), 5–39.

[2] Andersen, T. G., Bollerslev, T., Diebold, F. X. & Vega, C. (2003). Micro effects of macro announcements: Real-time price discovery in foreign exchange. *American Economic Review*, 93(1), 38–62.

[3] Andrews, D. W. K. (1991). Heteroskedasticity and autocorrelation consistent covariance matrix estimation. *Econometrica*, 59(3), 817–858.

[4] Bailey, D. H. & Lopez de Prado, M. (2014). The Deflated Sharpe Ratio: Correcting for selection bias, backtest overfitting and non-normality. *Journal of Portfolio Management*, 40(5), 94–107.

[5] Fama, E. F. & French, K. R. (1988). Business cycles and the behavior of metals prices. *Journal of Finance*, 43(5), 1075–1093.

[6] Grinold, R. C. (1989). The Fundamental Law of Active Management. *Journal of Portfolio Management*, 15(3), 30–37.

[7] Kunsch, H. R. (1989). The jackknife and the bootstrap for general stationary observations. *Annals of Statistics*, 17(3), 1217–1241.

[8] Ledoit, O. & Wolf, M. (2004). A well-conditioned estimator for large-dimensional covariance matrices. *Journal of Multivariate Analysis*, 88(2), 365–411.

[9] Lopez de Prado, M. (2018). *Advances in Financial Machine Learning*. Wiley.

[10] Muth, J. F. (1961). Rational expectations and the theory of price movements. *Econometrica*, 29(3), 315–335.

[11] Newey, W. K. & West, K. D. (1987). A simple, positive semi-definite, heteroskedasticity and autocorrelation consistent covariance matrix. *Econometrica*, 55(3), 703–708.

[12] Pearl, J. (2009). *Causality: Models, Reasoning and Inference* (2nd ed.). Cambridge University Press.

[13] Working, H. (1949). The theory of price of storage. *American Economic Review*, 39(6), 1254–1262.